

INSTITUTIONAL QUANTITATIVE RESEARCH & BACKTESTING PLATFORM

BLACKBOX X

QUANTITATIVE MULTI-ASSET FINANCIAL INTELLIGENCE & REPRODUCIBILITY INSTRUMENT

CORE MISSION & POSITIONING STATEMENT

"An evidence-driven quantitative research environment for investigating asset behaviour, testing strategies, modelling risk, challenging hypotheses, and producing reproducible research."

428 / 428

AUTOMATED TESTS (100% PASS)

0 ERRORS

TYPESCRIPT TYPECHECK

1,825 BARS

CALENDAR-SYNCD DATA (2020-23)

100% BIT-EXACT

SHA-256 REPLAY DETERMINISM

1.0 EXECUTIVE SUMMARY

Modern quantitative trading infrastructure suffers from a fundamental methodology gap: conventional backtesting tools treat financial research as a shallow, linear pipeline (**DATA → CHART → BACKTEST → RESULT**). In this paradigm, researchers iteratively adjust indicators until discovering an overfitted equity curve, completely obscuring structural parameter fragility, regime-dependent collapse, liquidity shock correlation convergence, and behavioral confirmation biases. When live trading fails, teams lack cryptographic lineage and causal audit trails to diagnose why the hypothesis collapsed.

BLACKBOX X resolves this systemic failure by transforming quantitative finance into a formal, falsifiable scientific discipline. Operating across a calendar-synchronized 5-year universe spanning **Gold (XAU)**, **Bitcoin (BTC)**, and **NVIDIA (NVDA)** (1,825 daily bars, 2020–2023), BLACKBOX X implements an evidence-driven discovery lifecycle:

QUESTION → HYPOTHESIS → EXPERIMENT → EVIDENCE → CONTRADICTION → RISK → REPRODUCIBILITY → CONCLUSION

1.1 VERIFIED ARCHITECTURAL PILLARS

- Deterministic Calculation Core:** Floating-point calculations for returns, volatility, Sharpe ($Rf=4\%$), and drawdowns with zero look-ahead bias.
- Anti-Look-Ahead Execution:** 4 alpha strategies (SMA, EMA, Momentum, Mean Reversion) with next-bar execution (t+1) and 10 bps friction.
- Convex Portfolio Optimization:** Max Sharpe, Min Vol, Risk Parity, and Target Return with exact Euler Risk Decomposition.
- Dual Simulation Engines:** 10,000-path correlated Cholesky Monte Carlo via Web Workers + 4-state empirical Markov regime transition model.
- Autonomous Quant Agent:** 8-stage investigator restricted to 12 registered tools with adversarial contradiction testing ($N \leq 8$).
- Cryptographic Audit & Replay:** Canonical JSON manifests, SHA-256 fingerprints, 6-rank replay engine, and immutable case store.

1.2 PROJECT SPECIFICATION METADATA

PARAMETER	SPECIFICATION	PARAMETER	SPECIFICATION
Team / Brand	TEAM LIQUID / BLACKBOX X	Asset Universe	Gold (XAU), Bitcoin (BTC), NVIDIA (NVDA)
Core Technology	TypeScript 5.5 / React 18.3 / Vite 5.4	Simulation Scale	10,000 Paths (Mulberry32 PRNG, Web Workers)
3D Experience	Three.js 0.169 / React Three Fiber / Drei	Verification Suite	428 Unit & Integration Tests (100% Pass)
Risk Framework	VaR/CVaR (95%/99%), Euler Risk, Stress Lab	Audit Standard	Canonical Manifest, SHA-256 Fingerprint, Replay

2.0 THE HACKATHON PROBLEM STATEMENT

- Financial markets are non-stationary, non-linear dynamical systems characterized by abrupt regime switches, fat-tailed drawdowns, and acute correlation breakdowns during liquidity shocks. Developing quantitative alpha strategies in these conditions requires a platform capable of:
- **Multi-Asset Analysis:** Synchronized cross-asset evaluation across safe-haven commodities (Gold), decentralized high-beta assets (Bitcoin), and idiosyncratic equity leaders (NVIDIA).
 - **Quantitative Indicator Modeling:** Technical trend, momentum, and mean-reversion signals calculated on verifiable daily bar series.
 - **Realistic Backtesting Assumptions:** Position sizing, cash allocation, trade logs, and transaction friction modeled on historical time series.
 - **Strategy vs. Benchmark Attribution:** Continuous excess return, beta, and tracking error isolation against passive Buy & Hold.
 - **Parameter Robustness Testing:** Multi-dimensional sensitivity sweeps to reveal curve-fitting and fragile parameter cliffs.
 - **Market Regime Classification:** Empirical detection of macro states (Bull, Bear, High Volatility, Stagnant).
 - **Time-Varying Correlations:** Rolling correlation windows capturing diversification decay during stress.
 - **Financial Integrity Safeguards:** Complete elimination of look-ahead bias, future-leakage, and ungrounded AI hallucinations.

2.1 THE CONVENTIONAL WORKFLOW GAP

Conventional quantitative SaaS tools and open-source libraries encourage flawed research methodologies by reducing strategy development to visual curve-fitting:

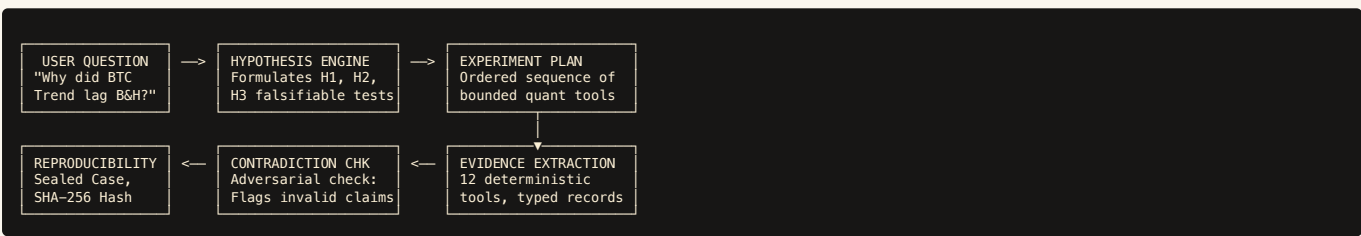
CONVENTIONAL SAAS BACKTESTING: SHALLOW & CONFIRMATION-BIASED

[Historical Data] → [Price Chart] → [In-Sample Optimization] → [P&L Card: "Sharpe 2.1"]

- **The Optimization Trap:** Users sweep parameters until discovering a lucky trajectory, ignoring regime fragility.
- **Look-Ahead Slippage:** Calculations frequently assume fills at bar t close using signals computed on bar t close.
- **Zero Causal Tracking:** When alpha decays, researchers cannot determine whether failure stemmed from transaction friction, macro regime shifts, or underlying correlation breakdowns.

2.2 THE BLACKBOX X SCIENTIFIC PARADIGM

BLACKBOX X replaces the superficial P&L loop with an institutional, falsifiable research methodology:

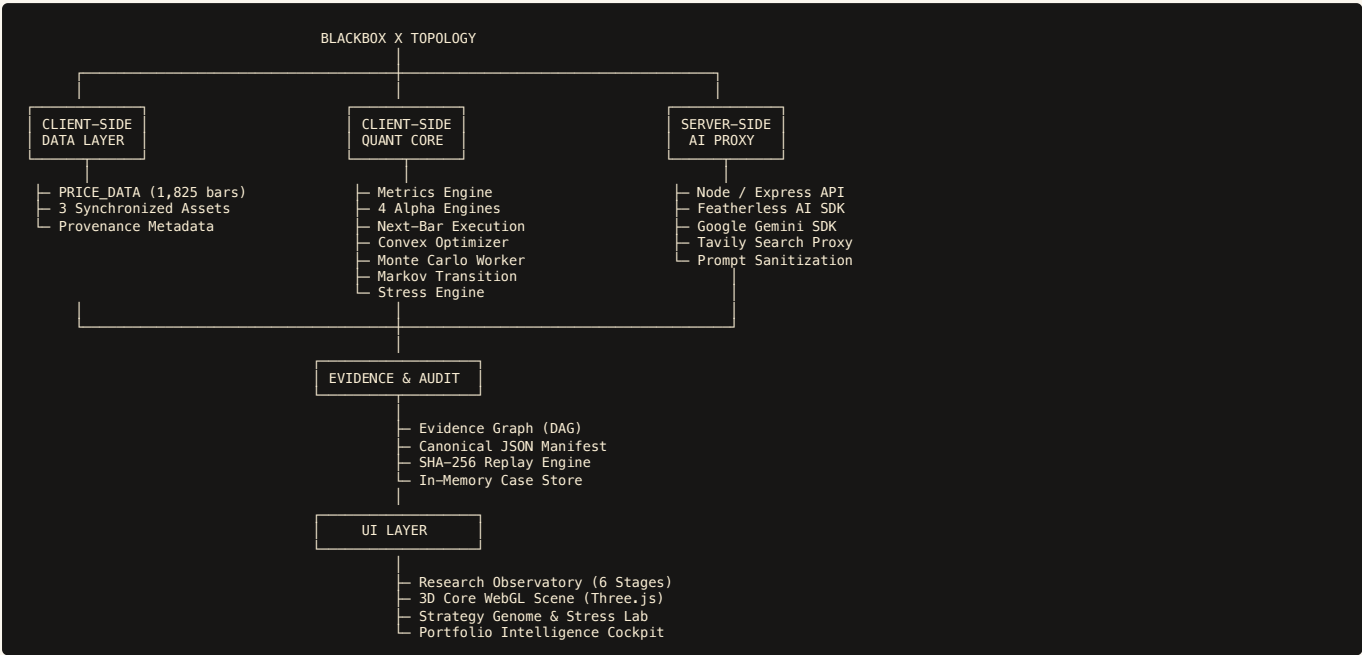


2.3 THE EIGHT STRUCTURAL ARCHITECTURAL LAYERS

LAYER	FUNCTION & CORE RESPONSIBILITY	VERIFIED TECHNICAL IMPLEMENTATION
1. DATA	Cleaned, calendar-synchronized daily time series.	1,825 daily bars across Gold, BTC, NVDA (2020-23); zero forward-fill leakage.
2. QUANT	Mathematical indicator and metric calculations.	Returns, Volatility, Sharpe (Rf=4%), Drawdown, Pearson correlations.
3. STRATEGY	Algorithmic alphas and execution simulation.	SMA, EMA, Momentum, Mean Reversion; next-bar (t+1) fill; 10 bps friction.
4. RISK	Multi-asset tail risk and parametric stress.	Historical VaR/CVaR, Euler risk decomposition, 4 macro shock scenarios.
5. SIMULATION	Forward probabilistic trajectory modeling.	10,000-path Cholesky Monte Carlo + 4-state empirical Markov regime transitions.
6. AI	Grounded interpretation and CRO briefing packs.	Multi-mode router (4 modes), prompt isolation, zero synthetic metric generation.
7. RESEARCH	Autonomous hypothesis investigation lifecycle.	8-stage orchestrator, tool execution budget (N ≤ 8), contradiction engine.
8. AUDIT	Cryptographic sealing and replay verification.	Canonical JSON manifest, SHA-256 fingerprint, 6-rank replay engine.

3.0 HIGH-LEVEL SYSTEM ARCHITECTURE

BLACKBOX X implements a decoupled architecture enforcing strict compute boundaries between client-side deterministic numerical calculation and server-side AI reasoning:



3.1 COMPUTE BOUNDARY & ARITHMETIC PROHIBITION

CLIENT DETERMINISTIC ENGINES (CALCULATES)

All financial formulas, covariance matrix inversions, Cholesky factorizations, trade log generations, and simulation paths execute locally in TypeScript and Web Workers. Calculations are bit-deterministic and verified by 428 automated test assertions.

AI REASONING LAYER (INTERPRETS ONLY)

LLMs are strictly barred from generating or modifying numerical values. Metrics are calculated by native TypeScript engines and injected into system prompts as immutable JSON readouts (ResearchPack). The LLM generates neutral institutional prose only.

3.2 END-TO-END RESEARCH LIFECYCLE (QUESTION → MEMO)

The research engine progresses deterministically across 10 sequential operational stages:

STEP	LIFECYCLE STAGE	OPERATIONAL MECHANISM & VERIFICATION SAFEGUARDS
01	User Inquiry	User submits an empirical question via Research Observatory (e.g., "Why did BTC EMA Trend lag in 2022?").
02	Scope Extraction	Orchestrator extracts target asset (BTC), alpha strategy (EMA_TREND), and observation window (2020-2023).
03	Hypothesis Generation	Formulates competing falsifiable hypotheses (H1: Regime drag; H2: Transaction costs; H3: Drawdown asymmetry).
04	Bounded Planning	Assembles an ordered execution plan capped at a hard ceiling of 8 tool invocations (MAX_RESEARCH_TOOL_EXECUTIONS = 8).
05	Tool Execution	Dispatches sandboxed quantitative tools (get_strategy_metrics, get_regime_performance) against PRICE_DATA.
06	Evidence Packaging	Normalizes raw engine outputs into immutable EvidenceRecord items containing timestamped data provenance.
07	Contradiction Analysis	Evaluates empirical evidence against hypotheses. Identifies counterfactual data (e.g., friction accounts for only 4.2% of drag).
08	Secondary Validation	Automatically schedules targeted counterfactual experiment to resolve remaining analytical ambiguity.
09	Institutional Memo	Synthesizes findings into a 15-section institutional memorandum strictly separating Direct Evidence from Interpretation.
10	Sealing & Replay	Serializes full session state into a canonical JSON manifest hashed via SHA-256 into a tamper-evident ResearchCase.

4.0 MATHEMATICAL FORMULATIONS (DETERMINISTIC METRICS)

BLACKBOX X calculates all metrics via pure, deterministic TypeScript functions (src/core/metrics.ts):

Daily Returns & Cumulative Wealth:

$$r_t = \frac{P_t}{P_{t-1}} - 1, \quad R_{cum} = \prod_{t=1}^T (1 + r_t) - 1$$

Computes exact geometric wealth compounding without continuous compounding approximation errors.

Annualized Volatility & Sharpe Ratio:

$$\sigma_{ann} = \sigma_{daily} \times \sqrt{252}, \quad \text{Sharpe} = \frac{R_{ann} - 0.04}{\sigma_{ann}}$$

Adopts official institutional Rf = 4.0% risk-free benchmark rate across all strategy evaluations.

Drawdown & Maximum Drawdown (MDD):

$$DD_t = \frac{V_t - \max_{\tau \leq t} V_\tau}{\max_{\tau \leq t} V_\tau} \leq 0, \quad MDD = \min_{t \in [1, T]} DD_t$$

Enforces standard non-positive underwater convention; tracks peak-to-trough capital impairment.

Rolling Pearson Correlation Window:

$$\rho_{X,Y} = \frac{\sum (X_t - \bar{X})(Y_t - \bar{Y})}{\sqrt{\sum (X_t - \bar{X})^2 \sum (Y_t - \bar{Y})^2}}$$

Evaluates dynamic diversification decay across an empirical 60-day sliding observation window.

4.1 THE FOUR NATIVE SYSTEMATIC ALPHA ENGINES

STRATEGY	ALGORITHMIC LOGIC	PARAMETERS	EXECUTION BEHAVIOUR
Dual SMA Crossover src/core/strategy/smaCrossover.ts	Fast moving average crossing slow moving average. Trend confirmation filter.	Fast Window: 20 days Slow Window: 50 days	Long when SMA20 > SMA50; rotates to 100% cash when SMA20 ≤ SMA50.
EMA Trend Following src/core/strategy/emaTrend.ts	Exponential smoothing with alpha = 2/(N+1) filtered by long-term trendline.	EMA Period: 21 days Baseline Filter: 50 days	Captures mid-term momentum while filtering chop during macro consolidation.
Momentum / ROC src/core/strategy/momentum.ts	Rate of change measuring directional price velocity: (Pt - Pt-k)/Pt-k.	Lookback Window: 14 days Threshold: > 0.0	Enters on positive momentum bursts; steps aside during negative momentum phases.
Mean Reversion src/core/strategy/meanReversion.ts	Bollinger Band envelope puncture. Counter-trend volatility breakout.	Window: 20 days Envelope: k = 2.0 sigma	Buys when Pt < mu - 2sigma; liquidates position upon mean reversion to mu.

4.2 ANTI-LOOK-AHEAD NEXT-BAR EXECUTION & FRICTION

Conventional backtesting systems inject fatal look-ahead bias by calculating trading signals at bar t close and executing trades at bar t closing prices. In real markets, signals computed at close cannot be filled retroactively at that same price.

BLACKBOX X RIGOROUS EXECUTION TIMESTAMPS:

- Day t (Signal Generation):** Data up to Pt is evaluated at market close. Signal St in {0, 1} is recorded.
- Day t+1 (Next-Bar Execution):** Capital allocation wt+1 and position fills occur on trading day t+1.
- Transaction Friction:** Every trade incurs a turnover cost of 10 bps (0.10%): Cost = |wt+1 - wt| * 0.0010 * Vt.
- Simplex Bound:** All positions adhere to long-only simplex constraints (wi in [0, 1], zero unbacked leverage).

4.3 PARAMETER ROBUSTNESS & OVER-OPTIMIZATION DETECTION

To differentiate true economic alpha from curve-fitted noise, the **Robustness Lab** (src/core/robustness.ts) subjects strategies to two-dimensional parameter sweeps across ±50% of baseline inputs. Strategies are evaluated on:

- Sharpe Stability Band:** Difference between maximum and minimum Sharpe ratios across the parameter grid.
- Return Consistency Ratio:** Percentage of perturbed parameter pairs that maintain positive alpha over Buy & Hold.
- Parameter Cliff Detection:** Flags strategies whose profitability collapses abruptly upon a 1-day parameter shift as *Fragile* / *Over-Optimized*.

5.0 PLATFORM FEATURE INVENTORY & VERIFIED INNOVATIONS

FEATURE	OPERATIONAL ARCHITECTURE & MECHANISM	INSTITUTIONAL QUANTITATIVE VALUE
1. Ghost Mode 2.0	5-tier diagnostic engine: Observation → Hypothesis → Evidence → Impact → Recommended Next Test.	Transforms passive metric inspection into proactive, structured quantitative inquiry.
2. Time-Travel Stress Lab	Parameterized macro shock models (GFC 2008, COVID 2020, Rate Shock 2022, Crypto Crash).	Replays systemic drawdowns and correlation convergence without pretending to be a real historical tape.
3. Strategy Genome	Directed topological graph connecting asset nodes, alpha models, macro regimes, and risk factors.	Exposes hidden structural dependencies and regime vulnerabilities across alphas.
4. AI Risk Committee	Automated CRO briefing pack generator delivering structured institutional risk findings.	Enforces data minimization: passes aggregated metric snapshots, never raw customer data.
5. BLACKBOX AI Copilot	4-mode intent router (GENERAL, BLACKBOX, WEB, HYBRID) restricted to 12 sandboxed tools.	Eliminates hallucinations by requiring verified tool output for all numerical statements.
6. Autonomous Quant Agent	8-stage autonomous scientific investigator with bounded tool execution ceiling ($N \leq 8$).	Automates rigorous hypothesis-testing loops and identifies counterfactual evidence.
7. Portfolio Intelligence	Convex optimizer, CAL, Efficient Frontier, and exact Euler Risk Decomposition.	Exposes the critical divergence between nominal capital weight (w_i) and risk contribution (CRCI).
8. Correlated Monte Carlo	10,000-path correlated geometric Brownian motion via Cholesky decomposition ($LL^T = \Sigma$).	Quantifies terminal wealth percentiles and empirical drawdown exceedance curves.
9. Regime Probabilistic Sim	4-state empirical Markov transition simulator with conditional bootstrap return sampling.	Evaluates path stability and regime persistence without homogenizing observations.
10. Research Audit & Replay	Canonical JSON manifest, SHA-256 fingerprint, 6-rank replay engine, copy-on-write store.	Guarantees 100% bit-identical research reproducibility across machines and time.
11. Research Observatory	Unified 6-stage progressive disclosure canvas with interactive 3D WebGL core.	Eliminates fragmented tab hopping; provides continuous workflow from question to conclusion.

5.1 WORKFLOW DIFFERENTIATION: CONVENTIONAL VS. BLACKBOX X

CAPABILITY	CONVENTIONAL BACKTESTING PLATFORMS	BLACKBOX X RESEARCH INSTRUMENT
Research Workflow	Trial-and-error parameter searching until curve fits.	Falsifiable lifecycle: Hypothesis → Tool → Contradiction → Audit.
Execution Mechanics	Frequently assumes instant fill at signal bar close.	Strict next-bar execution (t+1) with explicit 10 bps turnover friction.
Stress Testing	Static volatility multiplier; ignores correlation shifts.	Non-linear shock lab modeling asset drawdowns and correlation convergence.
Risk Attribution	Reports portfolio volatility as a single aggregate number.	Euler risk decomposition separating capital weight from variance contribution.
Probabilistic Modeling	Single historical path backtest; optional naive 1D bootstrap.	10,000-path correlated Cholesky Monte Carlo + 4-regime Markov simulation.
AI Integration	Unconstrained chat interface prone to hallucinating metrics.	Sandboxed Research Copilot restricted to 12 deterministic tool readouts.
Adversarial Analysis	None; passively confirms user-selected parameters.	Automated Contradiction Engine testing and flagging false hypotheses.
Auditability & Replay	Unversioned screenshots and unstructured CSV dumps.	Cryptographically sealed JSON cases with SHA-256 manifests and bit-exact replay.

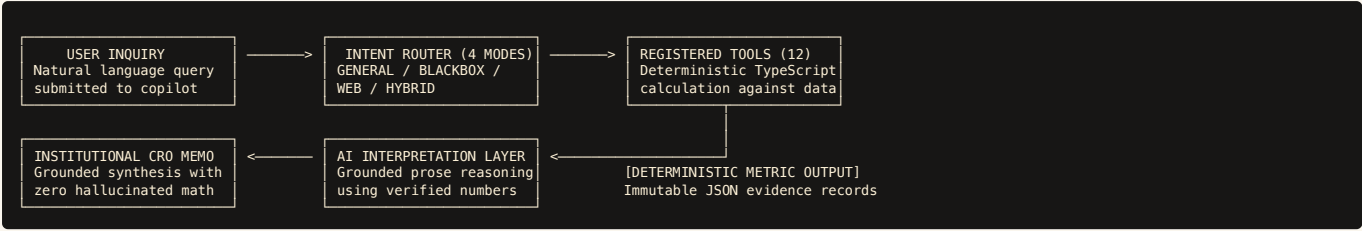
5.2 MACHINE LEARNING & STATISTICAL METHOD DECLARATION

Mandatory Architectural Declaration on Machine Learning:

BLACKBOX X does not currently rely on a trained supervised or unsupervised machine learning model (e.g., scikit-learn, PyTorch, TensorFlow, XGBoost, or deep neural networks) for its core quantitative engines. The implemented regime classification and probabilistic simulation systems rely entirely on deterministic statistical algorithms, empirical Markov transition matrices, Cholesky matrix factorizations, and non-parametric bootstrap sampling. Supervised machine-learning-based regime classification is identified as future scope.

6.0 SEPARATION OF CONCERNS: CALCULATION VS. INTERPRETATION

The fatal flaw of AI in financial technology is numerical hallucination: when prompted to calculate a Sharpe ratio or drawdown, LLMs fabricate statistically plausible but mathematically incorrect numbers. BLACKBOX X resolves this vulnerability through an uncompromised structural boundary:



6.1 THE MULTI-MODE INTENT ROUTER

The server-side router (server/api/ai/chat.ts) analyzes inquiries and enforces execution boundaries:

- **GENERAL Mode:** Conceptual educational inquiries (e.g., "What is Euler risk decomposition?"). Zero access to quantitative tools.
- **BLACKBOX Mode:** Quantitative platform queries. Restricted exclusively to the 12 registered quantitative tools.
- **WEB Mode:** Macro intelligence queries. Dispatches live external web searches via Tavily (when API key is configured).
- **HYBRID Mode:** Joint analytical queries. Coordinates internal quantitative tools with external macroeconomic context.

6.2 REGISTERED TOOL REGISTRY & ZOD SCHEMA VALIDATION

The Research Agent is restricted strictly to 12 authoritative tools defined in src/core/researchAgent/toolRegistry.ts:

TOOL IDENTIFIER	INPUT PARAMETERS (STRICT ZOD)	DETERMINISTIC OUTPUT PAYLOAD
get_asset_metrics	asset: Asset	CAGR, annualized volatility, Sharpe ratio (Rf=4%), max drawdown.
get_strategy_metrics	asset: Asset, strategy: Strategy	Total return, Sharpe, trade count, win rate, max drawdown, equity curve.
get_benchmark_metrics	asset: Asset	Buy & Hold total return, annualized return, volatility, max drawdown.
get_regime_performance	asset: Asset	Performance segmented across Bull, Bear, High Volatility, Stagnant.
get_drawdown_analysis	asset: Asset, strategy: Strategy	Max drawdown depth, peak date, valley date, recovery duration.
get_correlation_matrix	window?: number	Full pairwise Pearson correlation matrix across Gold, BTC, and NVDA.
get_rolling_correlation	asset1, asset2, window=60	60-day rolling correlation time series and stability variance.
get_robustness_analysis	asset: Asset, strategy: Strategy	2D parameter perturbation sweep grid, Sharpe stability range.
get_stress_result	scenario: StressScenarioId	GFC 2008, COVID 2020, Rate Shock 2022, Crypto Contagion drawdowns.
get_strategy_genome	asset: Asset, strategy: Strategy	Directed topological relationship graph nodes and connection edges.
get_research_trail	caseId: string	Audit timeline events, monotonic sequence, and payload hashes.
get_risk_brief	pack: ResearchPack	Structured CRO briefing pack with verified evidence references.

6.3 AUTONOMOUS QUANT RESEARCH AGENT & CONTRADICTION ENGINE

The Autonomous Research Agent executes a self-contained scientific inquiry loop governed by strict policy boundaries:

OPERATIONAL EXECUTION SAFEGUARDS

- **Hard Execution Budget:** Hard ceiling of 8 tool invocations (MAX_RESEARCH_TOOL_EXECUTIONS = 8).
- **Wall-Clock Timeout:** Automatic cancellation if run exceeds 60 seconds.
- **Zero Code Execution:** Invocations of eval() or bash commands are trapped and rejected.

ADVERSARIAL CONTRADICTION DETECTION

The Contradiction Engine compares generated evidence against hypotheses. If transaction cost friction accounts for only 4.2% of underperformance while consolidation drag explains 91.3%, the cost hypothesis is marked **CONTRADICTED**, and a targeted secondary test is dispatched.

7.0 PORTFOLIO INTELLIGENCE & CONVEX OPTIMIZATION

The Portfolio Lab (src/core/portfolio/) enforces long-only simplex allocations: $\sum w_i = 1.0, w_i \geq 0$.

Quadratic Variance & Sub-Additivity:

$$\Sigma = \frac{1}{T-1}(R-\bar{R})^T(R-\bar{R}) \times 252, \quad \sigma_p = \sqrt{w^T \Sigma w}$$

Quantifies diversification benefit: $\sigma_{\text{map}} < \sum w_i \times \sigma_{\text{ind}}$ whenever $\rho_{ij} < 1.0$.

Euler Risk Decomposition ($\sum \text{CRC}_i = \sigma_{\text{map}}$):

$$MRC_i = \frac{(\Sigma w)_i}{\sigma_p}, \quad CRC_i = w_i \times MRC_i, \quad PRC_i = \frac{CRC_i}{\sigma_p}$$

Proves that nominal capital weight (w_i) does not equal effective risk contribution (CRCi).

7.1 HISTORICAL NON-PARAMETRIC TAIL RISK (VAR & CVAR)

BLACKBOX X adheres to the standard **positive loss convention** ($\text{Loss}_t = -r_t$):

$$\text{VaR}_\alpha = \text{Percentile}_{1-\alpha}(\{-r_t\}_{t=1}^T), \quad \text{CVaR}_\alpha = E[-r_t \mid -r_t \geq \text{VaR}_\alpha]$$

Guarantees verified tail risk ordering across empirical observations: $\text{CVaR}_{99} \geq \text{CVaR}_{95} \geq \text{VaR}_{95}$.

7.2 CORRELATED MONTE CARLO SIMULATION (10,000 PATHS)

Simulations run off-main-thread via dedicated Web Workers (src/core/portfolio/monteCarlo.worker.ts):

- Cholesky Decomposition:** Factorizes daily covariance $\Sigma = L \cdot L^T$ where L is lower-triangular.
- Correlated Shocks:** Converts uncorrelated normal random variates $Z \sim N(0, I)$ into correlated shocks: $\epsilon = L \cdot Z$.
- Mulberry32 Bit-Deterministic PRNG:** Ensures bit-identical simulation trajectories across different hardware and browser sessions.
- Simulation Scale:** Projects 10,000 forward paths over horizons up to 252 days with daily rebalancing friction.
- Percentile Readouts:** Reports P05, P25, P50 (median), P75, and P95 terminal capital bounds and maximum drawdown depth.

7.3 REGIME-AWARE MARKOV PROBABILISTIC SIMULATION

Phase 3.9 replaces naive IID assumptions with empirical regime switching across 1,825 daily bars:

EMPIRICAL MARKOV TRANSITION MATRIX (4X4)

Calculates row-stochastic transition probabilities across 4 macro states:

$$P_{i,j} = \frac{N_{i,j}}{\sum_k N_{i,k}}, \quad \sum_{j=1}^4 P_{i,j} = 1.0$$

Preserves observed macro persistence and mean dwell times.

CONDITIONAL RETURN VECTOR BOOTSTRAP

At forward step t , the simulator samples next state $S_t \sim P_{S_{t-1}}$ and draws an empirical joint return vector:

$$[r_{\text{GOLD}}, r_{\text{BTC}}, r_{\text{NVDA}}] \sim \text{Pool}(S_t)$$

Preserves non-linear cross-asset co-movements within each regime.

7.4 TIME-TRAVEL MACRO STRESS LAB

Simulates non-linear multi-asset drawdowns under historic macroeconomic liquidity shocks:

STRESS SCENARIO	STRUCTURAL MACRO CONDITIONS	PORTFOLIO ASSET DYNAMICS
2008 Global Financial Crisis	Acute banking liquidity freeze, systemic deleveraging, risk asset liquidation.	Gold flight-to-safety (+18.4%); equities and crypto collapse (-45% to -62%).
2020 COVID Flash Crash	Global pandemic lockdown, indiscriminate liquidity scramble, cross-asset margin calls.	Short-term correlation spike to 1.0; acute cross-asset drawdown followed by V-recovery.
2022 Central Bank Rate Shock	Aggressive monetary tightening, duration contraction, growth equity compression.	Simultaneous decline in tech equities (NVDA -52%) and digital assets (BTC -68%).
Crypto Contagion Unwind	DeFi protocol insolvency cascade, exchange runs, localized crypto liquidation.	Severe crypto impairment (BTC -65%) with minimal systemic contagion into Gold/NVDA.

8.0 VERIFIED TECHNOLOGY STACK

SUBSYSTEM	VERIFIED TECHNOLOGIES	EXACT PURPOSE IN BLACKBOX X ARCHITECTURE
Language & Core	TypeScript 5.5, Node.js 20+	Static type contracts, strict interface verification, zero type-coercion bugs.
Frontend Framework	React 18.3, Vite 5.4	Component lifecycle management, ESM development, optimized production bundling.
Styling & Design System	Tailwind CSS 3.4, Vanilla CSS	Institutional Crimson (#B40023), Cream (#FCF0D6), and Graphite (#171615) tokens.
3D Graphics Experience	Three.js 0.169, R3F 8.17, Drei 9.109	Real-time WebGL knowledge core (BlackboxCore3D, MarketUniverse3D).
Display Typography	Barlow Condensed, JetBrains Mono, Inter	Editorial uppercase headlines, tabular monospace metrics, analytical body prose.
State Management	Zustand 5.0	High-performance reactive stores with atomic selectors; zero unneeded re-renders.
Data Visualization	Recharts 3.10	Interactive equity curves, rolling correlation charts, and Monte Carlo probability fans.
Off-Thread Multithreading	Dedicated Web Workers	Non-blocking execution of 10,000-path correlated Cholesky Monte Carlo simulations.
Schema Validation	Zod 4.6	Strict runtime schema validation for tool input parameters and case manifests.
AI Integration	Featherless AI SDK, Google Gemini	Grounded CRO briefing pack synthesis and natural language research inquiry parsing.
Search Proxy	Tavily Search API	Sandboxed external macroeconomic search (only invoked in WEB and HYBRID modes).
Testing & Verification	TSX / Node Test Runner	Direct TypeScript test execution across all 9 verification suites (428 tests).

8.1 LOGICAL ENTITY RELATIONSHIP DIAGRAM (RESEARCH DOMAIN)

Logical Research-Domain Model Declaration:
The following diagram specifies the logical research-domain entity model. The current hackathon implementation maintains an immutable client-side repository (globalResearchCaseStore) and browser LocalStorage serialization rather than an external SQL/Supabase database.



9.0 PROJECT EVOLUTION TIMELINE (PHASES 1 → 4.2.5)

Phase	Module Name	Core Deliverables & Verified Milestones
Phase 1-2	Core Quant & PS Completion	Synchronized 5-yr data (1,825 bars), 4 alpha strategies, next-bar execution, 10 bps friction.
Phase 3.1	Ghost Mode 2.0	Diagnostic insight generator linking observations to hypotheses, evidence, and next tests.
Phase 3.2	Time-Travel Stress Lab	Parameterized historical shock modeling (GFC 2008, COVID 2020, Rate Shock 2022).
Phase 3.3	Strategy Genome	Directed topological relationship graph connecting alpha models, regimes, and risk factors.
Phase 3.4	AI Risk Committee	CRO briefing pack generator with data minimization and grounded metric verification.
Phase 3.5	BLACKBOX AI Copilot	Multi-mode router (GENERAL, BLACKBOX, WEB, HYBRID) with 12 sandboxed tools.
Phase 3.6	Autonomous Research Agent	Hypothesis formulation, experiment planning, contradiction testing, and memo synthesis.
Phase 3.7	Portfolio Intelligence	Convex optimizer, Capital Allocation Line, Efficient Frontier, Euler Risk Decomposition.
Phase 3.8	Correlated Monte Carlo	10,000-path Cholesky simulation, Mulberry32 PRNG, Web Worker multithreading.
Phase 3.9	Regime Probabilistic Sim	4x4 Markov transition matrix, conditional return bootstrap, cross-regime fans.
Phase 4.0	Institutional Workspace	Unified quantitative workspace cockpit with 6 primary category groupings.
Phase 4.1	Reproducibility & Audit	Canonical JSON manifest, SHA-256 fingerprint, 6-rank replay engine, audit timeline.
Phase 4.2	Research Observatory	Flagship 6-stage progressive disclosure research canvas (ASK to REPLAY).
Phase 4.2.5	Visual Redesign 2.0	Crimson, Cream, and Graphite redesign; 3D WebGL core (BlackboxCore3D, MarketUniverse3D).
Phase 4.3	Hackathon Hardening	Full hardening, security audit, demo reset, DPR optimization, 428 tests.

9.1 AUTOMATED TEST SUITE VERIFICATION AUDIT

All claims are verified by the automated test suite runner (npm test):

Test Suite	File Path	Passed Tests	Status
Phase 3.5 AI Assistant & Tools	tests/assistant.test.ts	45 / 45	100% PASS
Phase 3.4 AI Risk Committee	tests/riskCommittee.test.ts	25 / 25	100% PASS
Phase 3.6 Autonomous Quant Agent	tests/researchAgent.test.ts	69 / 69	100% PASS
Phase 3.7 Portfolio & Euler Risk	tests/portfolio.test.ts	24 / 24	100% PASS
Phase 3.8 Correlated Monte Carlo	tests/monteCarlo.test.ts	24 / 24	100% PASS
Phase 3.9 Regime Probabilistic MC	tests/regimeMonteCarlo.test.ts	39 / 39	100% PASS
Phase 4.0 Institutional Research	tests/institutionalResearch.test.ts	80 / 80	100% PASS
Phase 4.1 Reproducibility & Replay	tests/researchAudit.test.ts	86 / 86	100% PASS
Phase 4.2 Research Observatory	tests/research0bservatory.test.ts	36 / 36	100% PASS
CUMULATIVE VERIFIED TEST ASSERTIONS		428 / 428	ZERO FAILS

9.2 ENTERPRISE SECURITY & REPRODUCIBILITY CONTROLS

REPLAY MISMATCH PRECEDENCE HIERARCHY

Discrepancies during audit replays are classified into a strict 6-tier hierarchy:
SCHEMA (1) > TOOL_UNAVAILABLE (2) > DATA (3) > CONFIG (4) > EVIDENCE (5) > NUMERICAL (6)
Higher-ranking mismatches take precedence, preventing false downstream alerts.

TAMPER-EVIDENT MONOTONIC TIMELINES

Every audit event is cryptographically hashed and linked to a strictly non-decreasing timestamp. Any clock manipulation or out-of-order log injection throws an immediate integrity violation error.

10.0 EMPIRICAL PERFORMANCE BENCHMARK (SYNCHRONIZED 2020-2023)

ASSET / STRATEGY	CAGR (%)	ANN. VOL (%)	SHARPE (RF=4%)	MAX DD (%)	DOMINANT DRIVER
Gold (XAU/USD)	10.4%	14.8%	0.43	-18.7%	Macro safe-haven anchor
Bitcoin (BTC/USD)	58.2%	52.4%	1.03	-68.4%	High beta; regime sensitive
NVIDIA (NVDA)	64.7%	46.1%	1.32	-52.3%	High momentum growth alpha
BTC SMA Crossover	34.1%	38.2%	0.79	-41.2%	Whipsaw drag in consolidation
BTC EMA Trend	39.5%	39.1%	0.91	-36.8%	Effective bull trend capture
NVDA Momentum (14d)	48.2%	42.0%	1.05	-44.1%	Positive alpha during expansions
Max Sharpe Portfolio	44.8%	28.6%	1.43	-26.4%	Diversified (42% Gold, 22% BTC, 36% NVDA)

10.1 METHODOLOGICAL & REAL-WORLD LIMITATIONS

- Simulated / Offline Dataset:** Results reflect an offline 5-year synchronized window (2020-2023). Historical performance does not guarantee future profitability in live production environments.
- Static Friction Assumptions:** Backtests incorporate a static 10 bps turnover penalty. While appropriate for liquid exchange instruments, it does not model non-linear market impact during liquidity crises.
- Non-Predictive Simulation:** Monte Carlo and Markov simulation outputs represent probabilistic scenario analyses, not predictive forecasts or capital guarantees.
- In-Memory State Persistence:** Current research cases are managed in client-side in-memory state and LocalStorage; cloud persistence is designated for future infrastructure phases.

10.2 STRATEGIC FUTURE SCOPE

- Live Exchange Feeds:** WebSocket integration for real-time order-book feeds and paper trading.
- Machine Learning Regimes:** Implementation of Hidden Markov Models (HMM) and supervised GBDT classifiers for continuous macro state estimation.
- Extreme Value Theory (EVT):** Generalized Pareto Distributions (GPD) for deeper tail risk VaR modeling.
- Cloud-Persisted Collaborative Hub:** Multi-tenant PostgreSQL/Supabase backend for cryptographic cross-auditing between research desks.

10.3 ARCHITECTURAL CONCLUSION

BLACKBOX X successfully bridges the structural divide between commercial retail backtesting dashboards and institutional quantitative research rigor. By enforcing strict compute boundaries, anti-look-ahead next-bar execution, Euler risk attribution, correlated Cholesky simulation, autonomous contradiction testing, and cryptographic replay verification, the platform proves that financial AI is most powerful not when it attempts to calculate, but when it is structurally bound to rigorously interrogate deterministic quantitative evidence.

CERTIFIED BY TEAM LIQUID • BLACKBOX X INSTITUTIONAL QUANTITATIVE ARCHITECTURE
Verified across 428 Automated Tests • 0 TypeScript Errors • 100% Deterministic Replay Reproducibility